Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A string can withstand a tension of $25\ \mathrm{N}$. What is the greatest speed at which a body of mass $1\ \mathrm{kg}$ can be whirled in a horizontal circle using a $1\ \mathrm{m}$ length of a string?

(A) $2.5\ \mathrm{ms}^{-1}$

(B) $5.0\ \mathrm{ms}^{-1}$

(C) $7.5\ \mathrm{ms}^{-1}$

(D) $10\ \mathrm{ms}^{-1}$

Q.2

A car moves at a speed of $36\ \mathrm{km\ h}^{-1}$ on a level road. The coefficient of friction between the tyres and the road is 0.8. The car negotiates a curve of radius R. If $g = 10\ \mathrm{ms}^{-2}$, the car will skid (or slip) while negotiating the curve if value of R is

(A) $10\ \mathrm{m}$

(B) $13\ \mathrm{m}$

(C) $14\ \mathrm{m}$

(D) $16\ \mathrm{m}$

Q.3

A body of mass $0.5\ \mathrm{kg}$ is whirled in a vertical circle at an angular frequency of $10\mathrm{rads}^{-1}$. If the radius of the circle is $0.5\ \mathrm{m}$, what is the tension in the string when the body is at the top of the circle? Take $g = 10\ \mathrm{ms}^{-2}$

(A) $10\ \mathrm{N}$

(B) $20\ \mathrm{N}$

(C) $30\ \mathrm{N}$

(D) $40\ \mathrm{N}$

Q.4

A steel metre rod of mass $m = 1.5\ \mathrm{kg}$ and length $L = 1m$ rests with its upper end against a smooth vertical wall and its lower end on rough horizontal floor. What should be the minimum coefficient of friction between the ground and the rod so that it can be inclined at an angle of 30° with the floor without slipping?

(A) 0.88

(B) 0.98

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(C) 0.87

(D) 0.67

Q.5

The kinetic energy of a particle of mass m moving along a circle of radius r depends on the distance covered as $K = as^2$ where a is a constant. Find the centripetal and tangential forces and also the total force acting on the particle as function of s.

(A) $2as\sqrt{1+\left(\frac{s}{r}\right)^2}$

(B) $3as\sqrt{1+\left(\frac{s}{r}\right)^2}$

(C) $as\sqrt{1+\left(\frac{s}{r}\right)^2}$

(D) $7as\sqrt{1+\left(\frac{s}{r}\right)^2}$

Q.6

A coin of mass $m = 10\ \text{g}$ is placed at a distance of $30\ \text{cm}$ from the centre of a disc. The disc is rotated at 30 rev/min about a vertical axis passing through its centre. What should be the minimum value of the coefficient of friction between the coin and the disc so that the coin does not skid off the disc?

(A) 0.1

(B) 0.2

(C) 0.3

(D) 0.4

Q.7

A conical pendulum has a string of length $l = 50\ \text{cm}$ and bob of mass $m = 200\ \text{g}$. The bob is revolved in a horizontal circle of radius $r = 20\ \text{cm}$. If the string makes an angle $\theta = 60^\circ$ with the vertical, find the tension in the string. Take $g = 10\ \text{ms}^{-2}$.

(A) 1 N

(B) 2 N

(C) 3 N

(D) 4 N

Q.8

A liquid of mass M and density ρ is filled in a tube AB of length L. The tube is rotated about end A with angular velocity ω. Obtain the expression for the force exerted at the other end B.

(A) $(\frac{1}{2}ML\omega^2$

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(B) $ML\omega^2$

(C) ML_ω

(D) $\frac{ML\omega^2}{4}$

Q.9

A body of mass $m = 20\ \text{g}$ is attached to an elastic spring of length $L = 50\ \text{cm}$ and spring constant $k = 2\text{Nm}^{-1}$. The system is revolved in a horizontal plane with a frequency $v = 30\text{rev/min}$. Find the radius of the circular motion and the tension in the spring.

(A) 0.05 m, 0.1 N

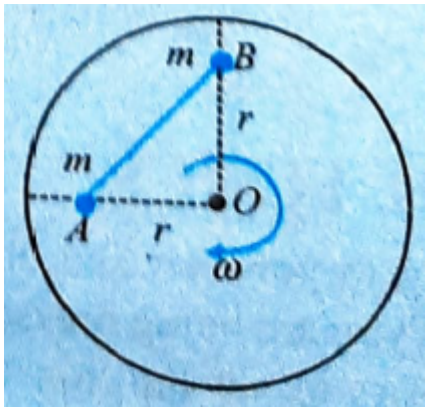
(B) 0.06 m, 0.2 N

(C) 0.075 m, 0.2 N

(D) 0.015 m, 0.3 N

Q.10

Figure shows top view of a circular rotating table, rotating with speed ω. Two particles connected by a string are kept on two mutually perpendicular radii. Coefficient of friction is μ. What can be the maximum angular speed of the table so that particles do not slip on it?



(A) $\sqrt{\frac{\sqrt{2}\mu g}{r}}$

(B) $\sqrt{\frac{\sqrt{5}\mu g}{r}}$

(C) $\sqrt{\frac{\sqrt{6}\mu g}{r}}$

(D) $\sqrt{\frac{\sqrt{7}\mu g}{r}}$

Q.11

A block of mass m is kept on rough horizontal turn table at a distance r from center of table. Coefficient of friction between turn table and block is μ. Now turn table starts rotating with uniform angular acceleration α. Find the time after which slipping occurs between block and turn table.

(A) $t = \left[\left(\frac{\mu g}{\alpha^2 r}\right)^2 - \left(\frac{1}{\alpha}\right)^2\right]^{1/4}$

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(B) $t=\left[\left(\frac{\mu g}{\alpha^2 r}\right)^2-\left(\frac{2}{\alpha}\right)^2\right]^{1/4}$

(C) $t=\left[\left(\frac{\mu g}{\alpha^2 r}\right)^2-\left(\frac{3}{\alpha}\right)^2\right]^{1/4}$

(D) $t=\left[\left(\frac{\mu g}{\alpha^2 r}\right)^2-\left(\frac{4}{\alpha}\right)^2\right]^{1/4}$

Q.12

A particle is moving on a circular path of radius $\frac{100}{\sqrt{19}}\ \mathrm{m}$ in such a way that magnitude of its velocity varies with time as $v=2t^2+t$, where v is velocity in $\mathrm{m/s}$ and t is time in s. The acceleration of the particle at $t=2\ \mathrm{s}$ is

(A) $21\ \mathrm{m/s^2}$

(B) $9\ \mathrm{m/s^2}$

(C) $10\ \mathrm{m/s^2}$

(D) $13.5\ \mathrm{m/s^2}$

Q.13

A stone tied to the end of a string of $1\ \mathrm{m}$ long is whirled in a horizontal circle with a constant speed. If the stone makes 22 revolutions in 44 seconds, what is the magnitude and direction of acceleration of the stone?

(A) $\pi^2\ \mathrm{m\ s^{-2}}$ and direction along the radius towards the centre

(B) $\pi^2\ \mathrm{m\ s^{-2}}$ and direction along the radius away from the centre

(C) $\pi^2\ \mathrm{m\ s^{-2}}$ and direction along the tangent to the circle

(D) $\pi^2/4\ \mathrm{m\ s^{-2}}$ and direction along the radius towards the centre

Q.14

A man running along a straight road with uniform velocity $\vec{\mathbf{u}}=u\hat{\mathbf{i}}$ feels that the rain is falling vertically down along $-\hat{\mathbf{j}}$ If he doubles his speed, he finds that the rain is coming at an angle θ with the vertical. The velocity of the rain with respect to the ground is

(A) $ui-uj$

(B) $ui-\frac{u}{\tan\theta}\hat{j}$

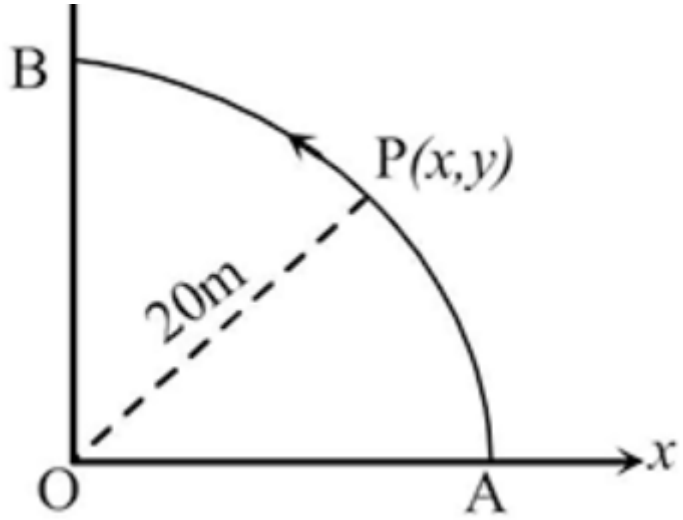
(C) $2u\hat{i}+u\cot\theta\hat{j}$

(D) $ui+u\sin\theta\hat{j}$

Q.15



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A point P moves in counter-clockwise direction on a circular path as shown in the figure. The movement of ' P ' is such that it sweeps out a length $s = t^3 + 5$, where s is in metres and t is in seconds. The radius of the path is $20\ \mathrm{m}$. Then what is the value of acceleration (in $\mathrm{m/sec^2}$) of ' P ' at $t = 2\ \mathrm{s}$?

(A) 13

(B) 12

(C) 16

(D) 14

Answers & Solutions

Q.1 Answer:

$5.0\ \mathrm{ms^{-1}}$

Solution:

Since the body is whirled in a horizontal circle, the gravity, acting vertically downwards, has no effect on the motion. If v is the greatest speed with which the body can be whirled, the maximum centripetal force (or tension) in the string is mv^2/R, which must balance a force of $25\ \mathrm{N}$. Thus

$25 = \frac{mv^2}{R} = \frac{1 \times v^2}{1}$

which gives $v = 5\ \mathrm{ms^{-1}}$, which is choice (b).

Q.2 Answer:

$10\ \mathrm{m}$

Solution:

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Speed of car $(v) = 36 \text{ km h}^{-1} = 10 \text{ ms}^{-1}$**. The maximum centripetal force that friction can provide is**

$$f_{\max} = \mu mg = \frac{mv^2}{R}$$

$$\text{or } R_{\min} = \frac{v^2}{\mu g} = \frac{10 \times 10}{0.8 \times 10} = 12.5 \text{ m}$$

This is the minimum radius the curve must have for the car to negotiate it without sliding at a speed of 10 ms^{-1}**. Hence the correct choice is (a).**

Q.3 Answer:

20 N

Solution:

It is clear from Fig., that when the body at the top point A **of the circle, its weight** mg **and tension** T_1 **in the string act downwards towards the centre** O **of the circle and the sum of the two provides the necessary centripetal force. Thus**

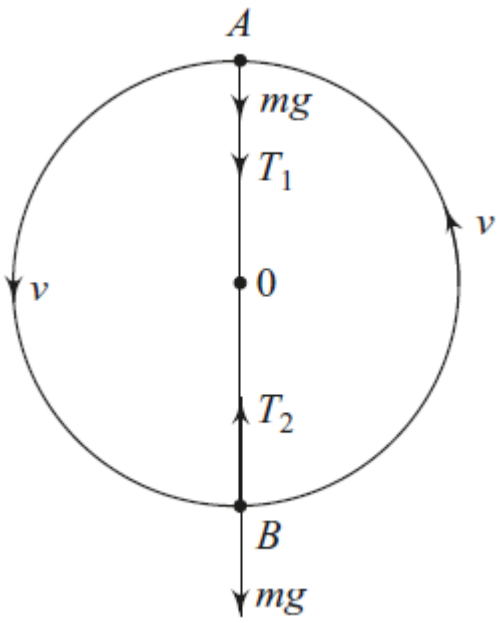
$$T_1 + mg = mR\omega^2$$

$$\text{Or } T_1 = m\left(R\omega^2 - g\right)$$

$$= 0.5 \times \left(0.5 \times 10^2 - 10\right)$$

$$= 20 \text{ N}$$

Thus, the correct choice is (b)



Q.4 Answer:

0.87

Solution:

Refer to Fig. Length of rod $AB = 1 \text{ m}$

N_1 **= normal reaction of the wall**

N_2 **= normal reaction of the floor**

The frictional force (f) **between the rod and the floor acts along** AD**. The weight** mg **of the rod acts at its midpoint (centre of mass)** C **so that** $AC = AB/2$**. For translational equilibrium,**

$$N_2 = mg \quad \text{and} \quad N_1 = f$$

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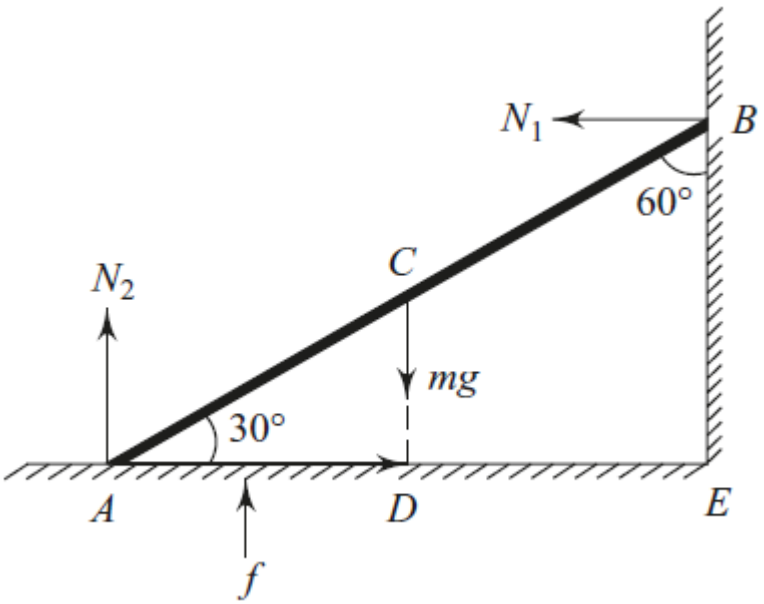
Taking moments of forces about A, we have for rotational equilibrium

$$N_2 \times 0 - mg \times AD + N_1 \times BE = 0$$

$$\Rightarrow \quad 0 - N_2 \times AD + f \times BE = 0 \, (\because mg = N_2, N_1 = f)$$

$$\Rightarrow \quad \frac{f}{N_2} = \frac{AD}{BE} = \frac{AC\cos 30^\circ}{AB \sin 30^\circ}$$

$$\Rightarrow \quad \mu = \frac{1}{2} \times \frac{\sqrt{3}/2}{1/2} = \frac{\sqrt{3}}{2} = 0.87$$



Q.5 Answer:

$2as\sqrt{1+\left(\frac{s}{r}\right)^2}$

Solution:

The kinetic energy of a particle, $K = \frac{1}{2}mv^2 = as^2$

or $v^2 = \frac{2as^2}{m}$ **...(i)**

Centripetal force, $F_C = \frac{mv^2}{r} = \frac{m}{r}\left(\frac{2as^2}{m}\right) = \frac{2as^2}{r}$ **...(ii)**

Tangential acceleration, $a_t = \frac{dv}{dt} = \frac{d}{dt}\left(\sqrt{\frac{2a}{m}}\right)s = \left(\sqrt{\frac{2a}{m}}\right)\frac{ds}{dt}$

$\Rightarrow a_t = \sqrt{\frac{2a}{m}}v = \sqrt{\frac{2a}{m}}\left(\sqrt{\frac{2a}{m}}s\right) = \frac{2as}{m}$

Tangential force, $F_t = ma_t = m\left(\frac{2as}{m}\right) = 2as$ **...(iii)**

Total force, $F = \sqrt{F_t^2 + F_c^2} = 2as\sqrt{1+\left(\frac{s}{r}\right)^2}$

Q.6 Answer:

0.3

Solution:

For no slipping, frictional force > centripetal force, i.e.

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$$\mu mg > m\omega^2 r$$

$$\Rightarrow \quad \mu > \frac{\omega^2 r}{g}$$

$$\Rightarrow \quad \mu > \frac{(2\pi v)^2 r}{g}$$

$$v = 30\text{rev/min} = \frac{30}{60} = 0.5 \text{ Hz},$$

$$r = 0.3 \text{ m and } g = 9.8 \text{ ms}^{-2}.$$

$$\therefore \quad \mu_{\min} = \frac{4\pi^2 v^2 r}{g}$$

$$= \frac{4 \times (3.14)^2 \times (0.5)^2 \times 0.3}{9.8}$$

$$= 0.3$$

Q.7 Answer:

4 N

Solution:

Refer to Fig.

$OQ = r$ **and** $PQ = 1$

Force towards centre $O = T \sin\theta$

i.e. $\frac{mv^2}{r} = T \sin\theta$ **(i)**

Also $mg = T \cos\theta$ **(ii)**

(a) From (ii) $T = \frac{mg}{\cos\theta} = \frac{0.2 \times 10}{\cos 60^\circ} = 4 \text{ N}$

Q.8 Answer:

$(\frac{1}{2} ML\omega^2$

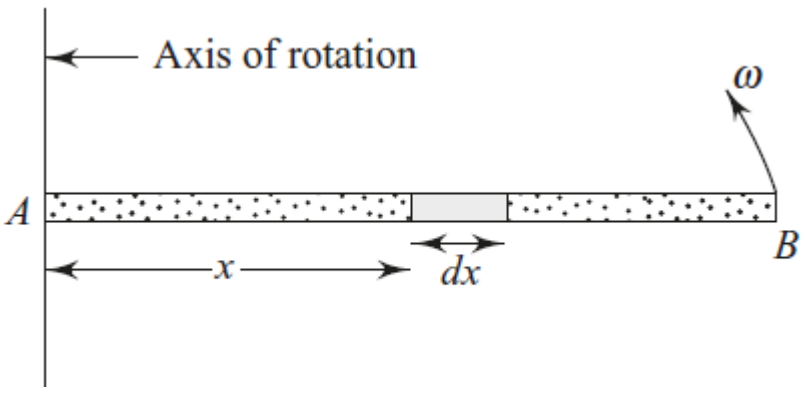
Solution:

Mass of element of length dx **is** $m = \frac{M}{L} dx$

$$\text{Centripetal force on element} = \int_o^L m\omega^2 x$$

$$= \frac{M\omega^2}{L} \int_o^L x dx$$

$$= \frac{1}{2} ML\omega^2$$



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Q.9 Answer:

0.05 m, 0.1 N

Solution:

Angular velocity

$$\omega = 2\pi v = 2\pi \times \frac{30}{60}$$
$$= \pi \text{rads}^{-1}$$

For an elastic spring force $F = kx$ **where** x **is the extension.**

Radius of circular motion $r = L + x$**. Centripetal force** $= mr\omega^2 = F$

Or $m(L+x)\omega^2 = kx$

$$\Rightarrow \quad x = \frac{mL\omega^2}{k - m\omega^2}$$
$$= \frac{0.02 \times 0.5 \times (3.14)^2}{2 - 0.02 \times (3.14)^2}$$
$$\simeq 0.05 \text{ m}$$
$$\therefore \quad r = L + x = 0.5 + 0.05$$
$$= 0.55 \text{ m}$$

$$\text{Tension } F = kx = 2 \times 0.05 \simeq 0.1 \text{ N}$$

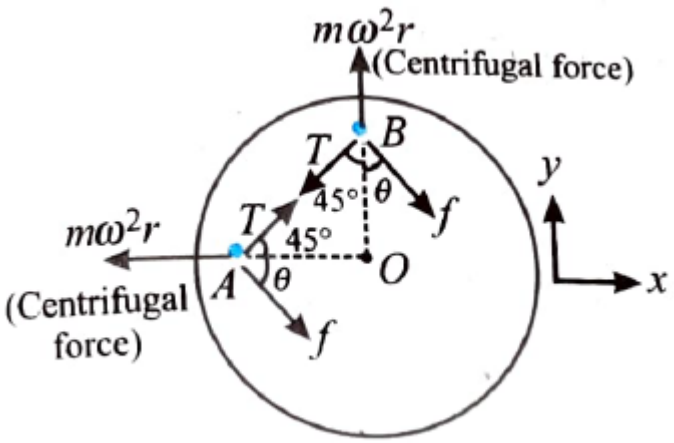
Q.10 Answer:

$\sqrt{\frac{\sqrt{2}\mu g}{r}}$

Solution:

We will solve this problem in the reference frame of table Here the centrifugal force on the particles will act in radial away as shown in free body diagrams of the blocks.

Friction force is static, therefore it is variable. Let friction act at an angle θ **with radial direction as shown in figure.** ${}_{c}a^{t}a^{2}$



In the impending state of motion, $f_{\max} = \mu N$

or particle A**,**

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In y-direction: $f\cos\theta + T\cos 45^\circ = mr\omega^2$ **...(i)**

In x-direction: $f\sin\theta = T\sin 45^\circ$ **...(ii)**

From Eqs. (i) and (ii) we eliminate T, to obtain

$mr\omega^2 = f(\sin\theta + \cos\theta)$ **...(iii)**

We can write Eq. (iii) as

$$mr\omega^2 = f\sqrt{2}\left(\frac{1}{\sqrt{2}}\sin\theta + \frac{1}{\sqrt{2}}\cos\theta\right)$$
$$= f\sqrt{2}\left(\sin\theta\cos 45^\circ + \cos\theta\sin 45^\circ\right)$$
$$= \mu mg\sqrt{2}\sin\left(45^\circ + \theta\right)$$
$$\omega^2 = \frac{\sqrt{2}\mu g}{r}\sin\left(45^\circ + \theta\right)$$

Since maximum value of $\sin\left(45^\circ + \theta\right) = 1$

$\therefore \omega_{\max} = \sqrt{\frac{\sqrt{2}\mu g}{r}}$

Q.11 Answer:

$t = \left[\left(\frac{\mu g}{\alpha^2 r}\right)^2 - \left(\frac{1}{\alpha}\right)^2\right]^{1/4}$

Solution:

Sol. Tangential acceleration of the block, $a_t = \frac{dv}{dt} = \alpha r$

Speed of the particle after time t **:** $\int_0^v dv = (\alpha r)\int_0^! dt \Rightarrow v = \alpha r t$

Centripetal acceleration, $a_c = \frac{v^2}{r} = \alpha^2 r t^2$

Net acceleration $a_{\text{net}} = \sqrt{a_t^2 + a_c^2} = \sqrt{\alpha^2 r^2 + \alpha^4 r^2 t^4}$

Net force acting on the block, $F_n = ma_{\text{net}} = m\sqrt{\alpha^2 r^2 + \alpha^4 + it^4}$

If the block starts slipping, $f = \mu_s mg$

$m\sqrt{\alpha^2 r^2 + \alpha^4 r^2 t^4} = \mu_s mg \Rightarrow \left(\alpha^4 r^2 t^4\right) = \mu^2 g^2 - \alpha^2 r^2$

$t = \left(\frac{\mu^2 g^2 - \alpha^2 r^2}{\alpha^4 r^2}\right)^{1/4} \Rightarrow t = \left[\left(\frac{\mu g}{\alpha^2 r}\right)^2 - \left(\frac{1}{\alpha}\right)^2\right]^{14}$

Q.12 Answer:

$10\ \text{m/s}^2$

Solution:

$a_t = \frac{dv}{dt} = 4t + 1 = 9\ \text{m/s}^2, \quad a_r = \frac{v^2}{r} = \frac{100\sqrt{19}}{100} = \sqrt{19}\ \text{m/s}^2$

$a_{net} = \sqrt{9^2 + (\sqrt{19})^2} = \sqrt{100} = 10\ \text{m/s}^2$

Q.13 Answer:

$\pi^2\ \text{m s}^{-2}$ and direction along the radius towards the centre

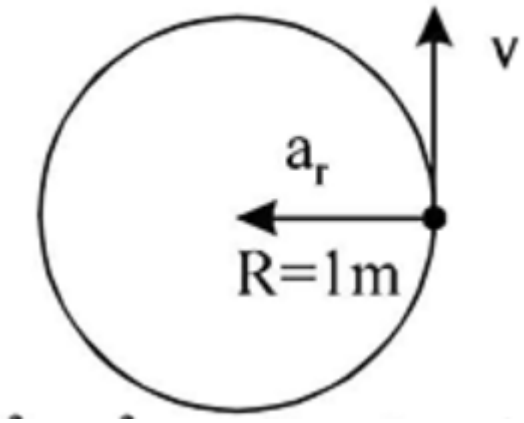
Solution:

$a_r = \omega^2 R \& a_t = \frac{dv}{dt} = 0$

or, $a_r = (2\pi n)^2 R = 4\pi^2 n^2 R = 4\pi^2\left(\frac{22}{44}\right)^2$

$a_{\text{net}} = a_r = \pi^2\ \text{ms}^{-2}$ **and direction along the radius towards the centre.**

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Q.14 Answer:

$ui - \frac{u}{\tan\theta}\hat{j}$

Solution:

Suppose velocity of rain $\overrightarrow{\mathbf{v}}_R = v_x\hat{\mathbf{i}} - v_y\hat{\mathbf{j}}$

and the velocity of the man $\overrightarrow{\mathbf{v}}_m = u\hat{\mathbf{i}}$

$\therefore$ **Velocity of rain relative to man** $\overrightarrow{\mathrm{v}}_{\mathrm{Rm}} = \overrightarrow{\mathrm{v}}_{\mathrm{R}} - \overrightarrow{\mathrm{v}}_{\mathrm{m}} = (\mathrm{v_x} - \mathrm{u})\,\hat{\mathrm{i}} - \mathrm{v_y}\hat{\mathrm{j}}$

According to given condition that rain appears to fall vertically, so $(v_x - u)$ **must be zero.** $\therefore$

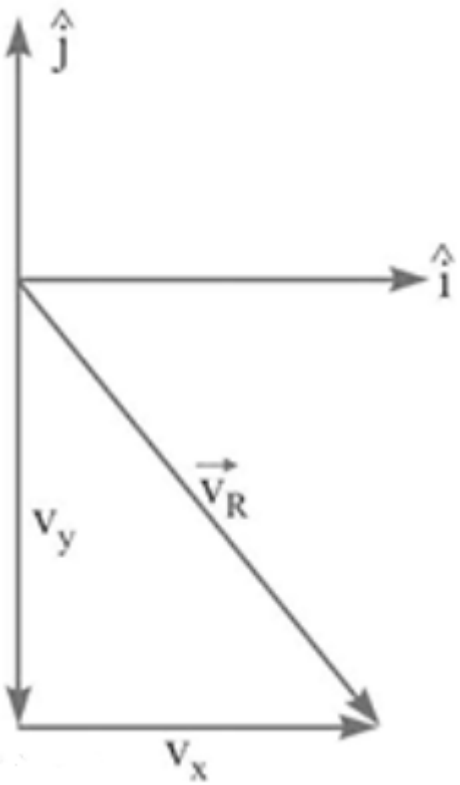
$v_{\mathrm{x}} - u = 0$ **or** $v_{\mathrm{x}} = u$

When he doubles his speed, $\overrightarrow{\mathbf{v}}_m = 2u\hat{\mathbf{i}}$

Now $\overrightarrow{\mathbf{v}}_{Rm} = \overrightarrow{\mathbf{v}}_R - \overrightarrow{\mathbf{v}}'_m$

$= \left(v_x\hat{\mathbf{i}} - v_y\hat{\mathbf{j}}\right) - (2u\hat{\mathbf{i}})$

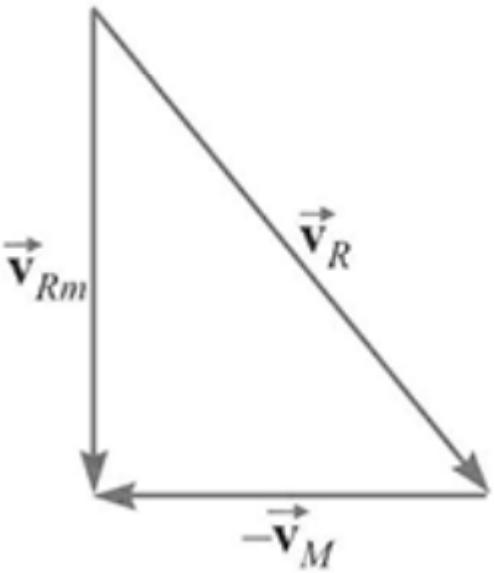
$= (v_x - 2u)\,\hat{\mathbf{i}} - v_y\hat{\mathbf{j}}$



The $\overrightarrow{\mathbf{v}}_{Rm}$ **makes an angle** θ **with the vertical** v_x

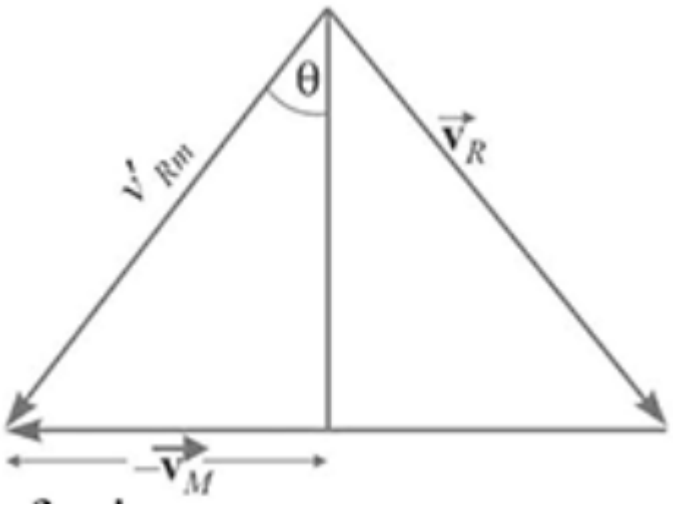
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$$\tan\theta = \frac{x-\text{ componend of } \mathbf{v}_{Rm}}{y-\text{ componend of } \vec{\mathbf{v}}_{Rm}}$$
$$= \frac{(v_x - 2u)}{-v_y}$$
$$= \frac{u-2u}{-v_y}$$



which gives

$$v_{\mathrm{y}} = \frac{u}{\tan\theta}$$



Thus the velocity of rain $\vec{\mathbf{v}}_R = v_x\hat{\mathbf{i}} - v_y\hat{\mathbf{i}}$
$= u\hat{\mathbf{i}} - \frac{u}{\tan\theta}\hat{\mathbf{j}}$

Q.15 Answer:

14

Solution:

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$= t^3 + 5$

$\Rightarrow$ velocity, $v = \dfrac{ds}{dt} = 3t^2$

Tangential acceleration $a_t = \frac{dv}{dt} = 6t$ **Radial acceleration** $a_c = \frac{v^2}{R} = \frac{9t^4}{R}$

At $t = 2s, \quad a_t = 6 \times 2 = 12\ \mathrm{m/s^2}$

$a_c = \frac{9 \times 16}{20} = 7.2\ \mathrm{m/s^2}$

$\therefore$ **Resultant acceleration**

$= \sqrt{a_t^2 + a_c^2} = \sqrt{(12)^2 + (7.2)^2} = \sqrt{144 + 51.84}$

$= \sqrt{195.84} = 14\ \mathrm{m/s^2}$

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